Microeconomics I

Class 5 — Indirect utility, wealth and price effects

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Plan for today

Warm-up; rice in Hunan

Indirect utility

Roy's identity

Break

Wealth effects: normal and inferior goods

Price effects: Giffen goods, substitutes, complements

In pairs: from v to demand; summary

Slide 1 Title

Preview version. The full slides and handout are posted after class.

Class 5 covers MWG 3.D, 3.G and 2.E: the value of the consumer's problem as a function of prices and wealth, the identity that recovers demand from it, and how demand moves when wealth, its own price or the other price changes. Class 4's best bundle is the input to everything today.

Slide 2 Plan for today

The morning in two questions about the best bundle found in class 4: how good it is, which is a number, the indirect utility $v(p, w)$; and how it moves when wealth and prices move. Before the break: v, its properties, and Roy's identity, which turns v back into demand. After the break: wealth effects, own-price effects and cross-price effects, each with a picture.

MWG 3.D, 3.G and 2.E; Muñoz-Garcia chapter 2 is the gentler read.

Practicalities

- Problem set 1: due at the start of class tomorrow, Wed 14 Oct, on paper, one copy per group
- Office hours: Wednesdays 14:00–15:00, Room 445

  up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu

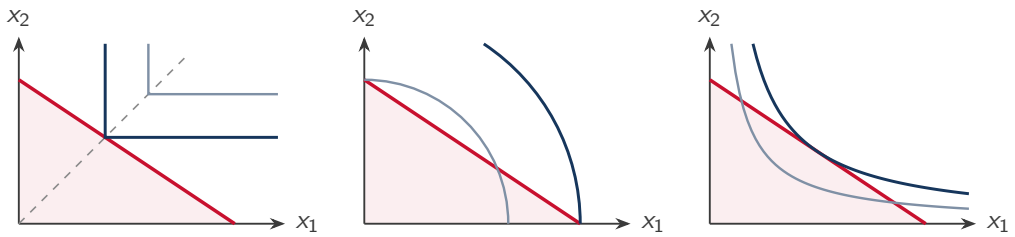
Slide 3 Practicalities

Problem set 1 is due at the start of class tomorrow, Wednesday 14 October, on paper, one copy per group, the answer sheet with all names on top of the working.

Office hours are on Wednesdays, 14:00–15:00, Room 445, up to 20 minutes per student; booking by e-mail gives priority. Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu.

Warm-up: where does each consumer buy?

$p = (2, 3)$, $w = 12$ in every panel



L-shaped, kinks on $x_2 = x_1$ **bowed out** $x_1 x_2$ **or** $\ln x_1 + \ln x_2$

Slide 4 Warm-up: where does each consumer buy?

Three consumers face class 4's budget, $p = (2, 3)$, $w = 12$, and the question is where each buys. Try it before class: for each panel decide whether a tangency exists at all and, if not, which point of the budget line wins. Panel C: do the two formulas draw different maps?

Answers revealed in class. One check from class 4: the tangency condition was derived for smooth curves and an interior bundle; look at what each panel does to that derivation.

Rice gets cheaper. Do poor households eat more rice?

Hunan, China, 2006: 650 poor urban households

rice: 64 % of their calories

at random, vouchers: up to a quarter off the price of rice, for five months

More rice, the same, or less?

A field experiment; what the households did, in class.

Slide 5 Rice gets cheaper. Do poor households eat more rice?

A real price cut for a staple. In 2006, 650 poor urban households in Hunan, China, got about 64 % of their calories from rice; at random, some of them received vouchers worth up to a quarter off the price of rice for five months, the rest were the comparison group. Did the households with cheaper rice eat more rice, the same, or less?

Decide before class and write down why. It is a question about how the demand for one good moves with its own price, the second half of today. The source of the experiment, and what the households did, in class.

Hunan: what the households did

Revealed in class

Before class: decide which answer you expect, and why.

Slide 6 Hunan: what the households did

Revealed in class.

Before class: decide which answer you expect, and why. Then ask what would have to be true of a household's spending for the opposite answer to come out.

Indirect utility

MWG 3.D, 3.G

slide 7 of 26

Slide 7 Indirect utility

Part 1: the value of the consumer's problem as a function of prices and wealth, and how to get demand back from it. MWG 3.D and 3.G.

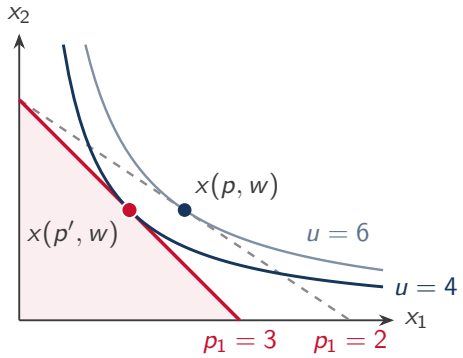
Indirect utility: the level of the best curve the budget reaches

$v(p, w) = u(x^*)$ for any $x^* \in x(p, w)$

the value of the UMP: $\max\{u(x) : x \in \mathcal{B}_{p,w}\}$

$u = x_1 x_2$: $\quad v(p, w) = \dfrac{w}{2p_1} \cdot \dfrac{w}{2p_2} = \dfrac{w^2}{4p_1 p_2}$

at $p = (2, 3)$: $w^2/24$ (class 4, slide 26)



p_1: $2 \to 3$: v falls from 6 to 4.

MWG 3.D, p. 56.

slide 8 of 26

Slide 8 Indirect utility: the level of the best curve the budget reaches

The indirect utility function assigns to each price–wealth pair the utility of the best bundle it affords: a number, not a bundle, and the same number whichever of several best bundles is used. Demand says where the best bundle is; v says how good it is. Both are functions of the three numbers p_1, p_2, w.

For $x_1 x_2$ the formula comes from substituting class 4's demand into u; at $p = (2, 3)$ it is the $w^2/24$ whose slope that class called λ. The picture is the definition read geometrically: a dearer good 1 pivots the budget line inward, the highest reachable indifference curve drops from 6 to 4, and that drop is the fall in v.

What any v must look like, whatever the preference, is the next slide.

$v(p, w)$ is homogeneous, rises with w and never rises with prices

property	why	for $v = \dfrac{w^2}{4p_1p_2}$
(i) homogeneous of degree 0	the same budget set	$\dfrac{(\alpha w)^2}{4\,\alpha p_1\,\alpha p_2} = v$
(ii) strictly increasing in w	LNS (class 2)	$\dfrac{\partial v}{\partial w} = \dfrac{w}{2p_1p_2} > 0$
nonincreasing in each p_ℓ	the budget set shrinks	$\dfrac{\partial v}{\partial p_1} = -\dfrac{w^2}{4p_1^2p_2} < 0$
(iii) quasiconvex in (p, w)	next slide	
(iv) continuous		

MWG Proposition 3.D.3, p. 56: u continuous, $\succsim$ locally non-satiated, $X = \mathbb{R}^2_+$.

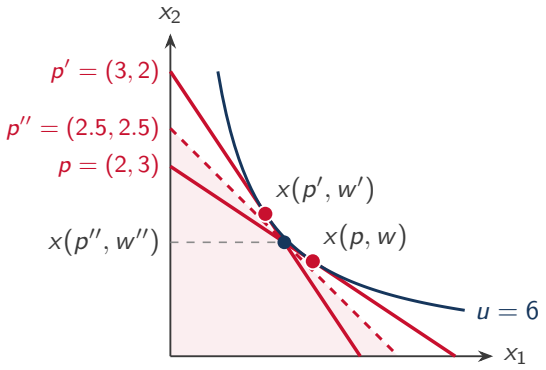
Slide 9 $v(p, w)$ is homogeneous, rises with w and never rises with prices

Proposition 3.D.3 lists what every indirect utility function satisfies, under a continuous u and local non-satiation. Homogeneity of degree zero is inherited from the budget set: scaling prices and wealth together changes nothing, so no assumption on u is needed. Strictly increasing in wealth needs local non-satiation: extra wealth leaves the old best bundle with money to spare, and a strictly better bundle sits close by, still affordable; with a bliss point inside the budget set extra wealth would be worthless.

In prices only nonincreasing, not decreasing: a higher price shrinks the budget set, but a good the consumer does not buy can get dearer without hurting her (class 4's corner solutions). The third column checks each property on the Cobb–Douglas v by hand. Continuity MWG states without proof.

Property (iii), quasiconvexity, is the one that surprises: next slide.

v is quasiconvex: the average of two budgets never beats both



$\{(p, w) : v(p, w) \leq \bar{v}\}$ is convex for every $\bar{v}$

$p = (2, 3)$ or $p' = (3, 2)$, $w = 12$: $v = 6$ both

their average $(2.5, 2.5)$, $w = 12$: $v = 5.76$

What the average budget affords, one of the two already affords: $v \leq 6$.

MWG Proposition 3.D.3(iii), p. 56; Figure 3.D.5, p. 57, in Wednesday's reading; $u = x_1x_2$.

Slide 10 v is quasiconvex: the average of two budgets never beats both

Quasiconvex means: for every level $\bar{v}$, the set of price–wealth pairs at which $v \leq \bar{v}$ is convex. Take two budgets at which $v \leq \bar{v}$ and average them. A bundle affordable at the averaged budget satisfies a weighted sum of the two original budget inequalities, so it satisfies at least one of them: it was already affordable at one of the two budgets, hence gives at most $\bar{v}$. In the picture the dashed averaged line lies inside the union of the two budget sets, and all three lines pass through $(2.4, 2.4)$.

This is quasiconvexity, not quasiconcavity, and nothing about the shape of u was used. Class 3's quasiconcave u had convex upper contour sets in the space of bundles; here the lower contour sets of v, in the space of prices and wealth, are convex. The intuition: extreme price vectors let the consumer specialise in whatever is cheap, so the averaged prices can never beat both originals. The numbers on the slide are the Cobb–Douglas check.

In pairs: could this be an indirect utility function?

Test (i) and (ii) of Prop. 3.D.3 (slide 9). Four minutes.

$v(p, w) =$	homogeneous of degree 0	$\uparrow$ in w, $\downarrow$ in p
$\dfrac{w}{p_1 + p_2}$		
$\dfrac{w^2}{p_1 p_2}$		
$\dfrac{p_1 + p_2}{w}$		
$\ln w - \ln p_1 - \ln p_2$		

Answers revealed in class. Try it in pairs before class.

Slide 11 In pairs: could this be an indirect utility function?

Try it before class. Two tests from Proposition 3.D.3 (slide 9), each a necessary property of any indirect utility function under local non-satiation. Homogeneity: replace (p, w) by $(\alpha p, \alpha w)$, prices and wealth together, and see whether α cancels. Direction: the signs of $\partial v/\partial w$ and $\partial v/\partial p_\ell$. Write a reason for every cell and, for a formula that passes, try to name the consumer it belongs to.

Answers revealed in class. Two common slips: checking only the directions, and putting α into the prices alone.

Change the utility function: v and λ change, demand does not

u	$v(p, w)$	$\lambda = \partial v/\partial w$ at $p = (2, 3)$, $w = 12$
$x_1 x_2$	$\dfrac{w^2}{4 p_1 p_2}$	$\dfrac{w}{2 p_1 p_2} = 1$
$\frac{1}{2} \ln x_1 + \frac{1}{2} \ln x_2$	$\ln w - \frac{1}{2} \ln p_1 - \frac{1}{2} \ln p_2 - \ln 2$	$\dfrac{1}{w} = \dfrac{1}{12}$

Same $\succsim$, same $x(p, w) = (3, 2)$; v becomes $f(v)$, and λ with it.

MWG p. 57, in Wednesday's reading ($\tilde{u} = f(u)$ gives $\tilde{v} = f(v)$; Example 3.D.2 with $\alpha = \frac{1}{2}$); $\lambda = \partial v/\partial w$: pp. 54–55, class 4, slide 26.

Slide 12 Change the utility function: v and λ change, demand does not

Relabel utility by a strictly increasing f and the indirect utility becomes $f(v)$: the best bundle is the same, so its utility is just renamed. The second row is Example 3.D.2 with $\alpha = \frac{1}{2}$, the log form of Cobb–Douglas, where $f(t) = \frac{1}{2} \ln t$.

Class 4's λ, the marginal utility of wealth, now has a home: it is the slope of v in w, by the chain rule and Walras' law. It is rescaled by f' along with v; here $f'(6) = 1/12$ turns 1 into $1/12$. So v and λ are properties of the representation; the bundle is a property of the preference: class 3's ordinal versus cardinal distinction again.

If demand survives every f while v does not, there should be a way to get demand out of v that does not care about f: Roy's identity, next.

Roy's identity: demand from two slopes of v

$$x_\ell(p,w) = -\frac{\partial v(p,w)/\partial p_\ell}{\partial v(p,w)/\partial w}$$

u continuous; $\succsim$ locally non-satiated and strictly convex; v differentiable at $(p,w) \gg 0$

$$v = \frac{w^2}{4p_1p_2}: \quad -\frac{-w^2/(4p_1^2p_2)}{w/(2p_1p_2)} = \frac{w}{2p_1}$$

Cobb–Douglas, class 4

Replace u by $f(u)$: both slopes gain the factor f', and the ratio does not change.

MWG Proposition 3.G.4 (Roy's identity), p. 73.

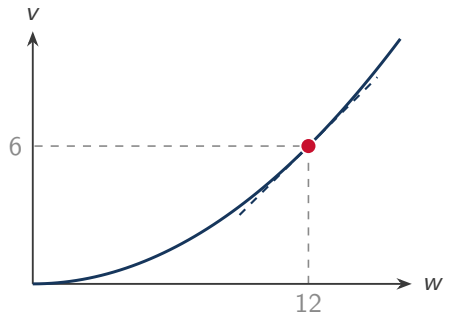
Slide 13 Roy's identity: demand from two slopes of v

Roy's identity recovers each demand from two partial derivatives of v: minus the slope in that good's price, divided by the slope in wealth. The minus sign is there because a higher price lowers v while higher wealth raises it, so the quotient of the raw slopes is negative; a negative demand is the alarm that the sign was dropped.

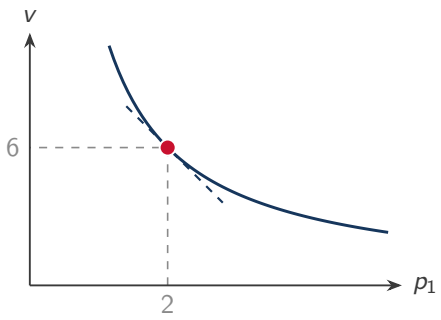
It does not care about the utility representation: replacing u by $f(u)$ multiplies both slopes by the same f', which cancels in the ratio. That is why demand survives relabelling while v and λ do not (slide 12). The assumptions matter: strictly convex preferences make demand a single bundle, and the warm-up's bowed-out consumer, whose demand jumps at $p_1 = 3$, has no Roy's identity there.

Why it is useful: demand by differentiation, with no system of first-order conditions to solve. The next slide says why the formula is true.

A dearer good costs utility in proportion to how much of it she buys



$p = (2,3)$: slope $\lambda = 1$



$w = 12$, $p_2 = 3$: slope $-\lambda x_1 = -3$

$$\frac{\partial v}{\partial w} = \lambda, \qquad \frac{\partial v}{\partial p_\ell} = -\lambda\, x_\ell(p,w)$$

Here $x_1 = -(-3)/1 = 3$: the bundle $(3,2)$ again.

MWG p. 74, Proofs 2 and 3 (the envelope theorem); $u = x_1x_2$, $p = (2,3)$, $w = 12$.

Slide 14 A dearer good costs utility in proportion to how much of it she buys

The two slopes have meanings of their own. A euro of extra wealth is worth λ in utility (left panel, the picture of class 4's slide 26, slope 1 at $w = 12$). A small rise in p_ℓ makes the same bundle cost x_ℓ euros more per unit of price rise, which is like losing x_ℓ euros of wealth; the consumer does re-optimise, but at an optimum that adjustment has no first-order effect on utility, the envelope theorem, so the loss is λx_ℓ per unit of price (right panel, slope -3 at $p_1 = 2$). Dividing the two slopes gives x_ℓ: Roy's identity.

MWG's second proof is the same argument in formulas: the chain rule gives $\partial v/\partial p_1$ as λ times $\sum_k p_k\, \partial x_k/\partial p_1$, and differentiating the budget identity $p \cdot x(p,w) = w$ in p_1 shows that this sum equals $-x_1$; slide 20 gives that identity a name. The misconception to drop: that the re-optimisation after a price change adds anything of first order.

(optional) A bliss point: does more wealth always raise v?

$u(x) = -(x_1 - 2)^2 - (x_2 - 2)^2$ on $\mathbb{R}^2_+$, $p = (2, 3)$.

(1) $v(p, w)$ for $w \geq 10$.

(2) Which part of Prop. 3.D.3 fails?
Why?

(3) $\lambda = \partial v / \partial w$ at $w = 12$?

At home if not done in class; the answers are in the slides released after class.

MWG Proposition 3.D.3, p. 56 (slide 9).

Slide 15 (optional) A bliss point: does more wealth always raise v?

An optional exercise, to be done at home if class time runs out: a consumer whose utility peaks at one bundle and falls off in every direction from it. Before touching the budget, find that bundle and ask from which wealth on she can afford it; then go down slide 9's list and ask which property her v breaks, and which assumption of Prop. 3.D.3 she violates.

The answers are in the slides and handout released after class. For (3), recall what λ measures (slide 12) before differentiating anything.

Break

Ten minutes

Slide 16 Break

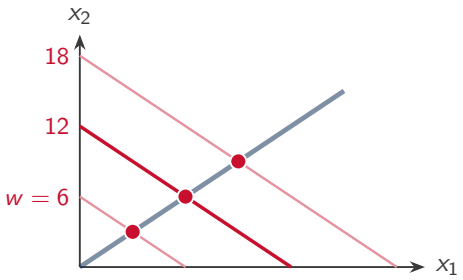
Ten minutes.

Wealth and price effects

MWG 2.E

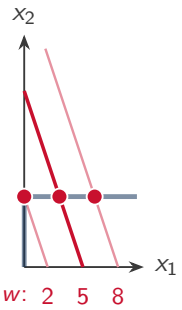
More wealth: Cobb–Douglas scales up; quasilinear: all to good 1

Engel function: $w \mapsto x(p, w)$, p fixed; wealth effect $\partial x_\ell / \partial w$



$u = x_1 x_2$, $p = (2, 3)$

$x = (w/4,\ w/6)$: a ray



$u = x_1 + \ln(1 + x_2)$, $p = (3, 1)$

$x_2 = 2$ once $w \geq 2$

Class 4's promise: past $w = 2$, every extra euro goes to good 1.

MWG 2.E, pp. 24–25, Figure 2.E.1 (the wealth expansion path).

Slide 17 Wealth and price effects

Part 2: the best bundle and its value are known; now wealth moves, then the good's own price, then the other price, and the bundle is watched. MWG 2.E.

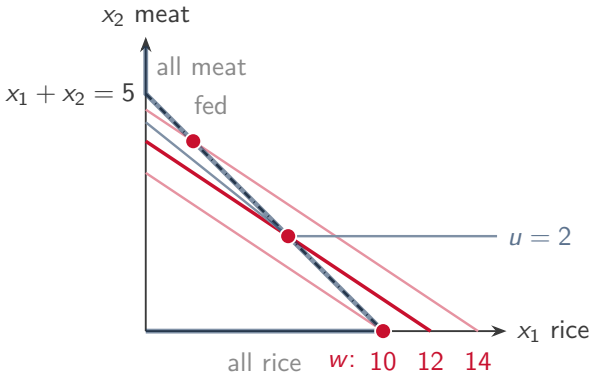
Slide 18 More wealth: Cobb–Douglas scales up; quasilinear: all to good 1

Fix prices and vary wealth: the Engel function; its trace in goods space is the wealth expansion path, MWG Figure 2.E.1 ("income" in other books). Cobb–Douglas spends fixed shares, so the bundle scales along a ray: both wealth effects positive, budget shares constant.

The right panel keeps class 4's promise for $x_1 + \ln(1 + x_2)$ at the prices of its Exercise 3, $p = (3, 1)$. Up to $w = 2$ she buys only good 2; from then on x_2 sits at $p_1/p_2 - 1 = 2$ and every extra euro goes to good 1, so class 4's rows 1 and 2 are two points on this path. Quasilinear utility kills the wealth effect on the nonlinear good once both are bought, which is why it is the workhorse for consumer surplus.

Can a wealth effect be negative? Next slide.

Rice can be inferior: past subsistence, more wealth buys less of it



$p = (2, 3)$

she needs 5 units of food:
$x_1 + x_2 \geq 5$

once fed, only meat counts

normal at (p, w): $\partial x_\ell / \partial w \geq 0$;
inferior: < 0

Rice is normal up to $w = 10$, inferior from 10 to 15.

$u = \min\{x_2,\ x_2 - 4(5 - x_1 - x_2)\}$: meat, minus 4 per unit of missing food. MWG 2.E, p. 25; own example.

Slide 19 Rice can be inferior: past subsistence, more wealth buys less of it

A consumer built for the Hunan question. Rice is the cheap food, meat the dear one; she needs five units of food, each missing unit costs her four utils, and once fed only meat counts. Her indifference curves kink on the food line, so while she can just afford to be fed, the best bundle sits on that line.

Three wealth zones. Below $w = 10$ she cannot reach five units, every euro goes to the cheapest food, and rice is normal. From 10 to 15 she is just fed, and each extra euro swaps one unit of rice for one of meat, which also feeds her, at a cost of $3 - 2 = 1$ euro: rice falls one for one with wealth, inferior, while meat is normal. Above 15 she is fed on meat alone and buys no rice.

Normal and inferior are properties at a given (p, w), not of a good forever; MWG notes that inferiority is common for narrow goods, through substitution toward higher quality as wealth rises. MWG counts a zero wealth effect as normal.

The budget ties the effects together: not every good can be inferior

$$\sum_k p_k \frac{\partial x_\ell}{\partial p_k} + w \frac{\partial x_\ell}{\partial w} = 0 \qquad \text{homogeneity: Prop. 2.E.1}$$

$$\sum_\ell p_\ell \frac{\partial x_\ell}{\partial w} = 1 \qquad \text{Walras in } w\text{: Engel aggregation, Prop. 2.E.3}$$

$$\sum_\ell p_\ell \frac{\partial x_\ell}{\partial p_k} + x_k = 0 \qquad \text{Walras in } p_k\text{: Cournot aggregation, Prop. 2.E.2}$$

Slide 19 at $w = 12$: $2 \cdot (-1) + 3 \cdot (+1) = 1$. An extra euro is spent somewhere.

MWG 2.E, pp. 27–28.

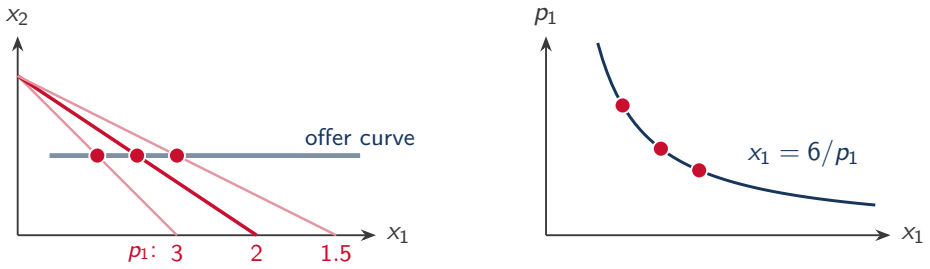
Slide 20 The budget ties the effects together: not every good can be inferior

Three identities that every Walrasian demand satisfies, each the derivative of something known. Differentiate homogeneity $x(\alpha p, \alpha w) = x(p, w)$ in α: line 1. Differentiate Walras' law $p \cdot x(p, w) = w$ in w: line 2, Engel aggregation; in p_k: line 3, Cournot aggregation, which was the step inside Roy's proof on slide 14.

Line 2 says an extra euro is spent somewhere: the price-weighted wealth effects add to one, so with two goods at most one can be inferior. The slide's check is slide 19's consumer at $w = 12$, where rice loses one unit and meat gains one. MWG also writes these identities with elasticities (Exercise 2.E.2).

Cheaper good 1: Cobb–Douglas buys more of it, as much of good 2

usual: $\partial x_\ell / \partial p_\ell \leq 0$; Giffen at (p, w): $\partial x_\ell / \partial p_\ell > 0$



goods: the offer curve **prices: the demand curve**

$x_1 = 6/p_1$ rises as p_1 falls; x_2 stays at 2: the offer curve is flat.

MWG 2.E, pp. 25–26, Figures 2.E.2–2.E.3; $u = x_1 x_2$, $w = 12$, $p_2 = 3$.

Slide 21 Cheaper good 1: Cobb–Douglas buys more of it, as much of good 2

The own-price effect, read two ways. Left, in goods space: as p_1 falls the budget line pivots outward around $(0, 4)$, and the best bundles trace the offer curve, MWG Figure 2.E.3. Cobb–Douglas spends half of wealth on good 1 whatever its price, so $x_1 = 6/p_1$ rises while x_2 stays at 2: class 4's "nothing happens to x_2" is the flat offer curve.

Right, the same information as a demand curve with the price on the vertical axis, the economists' habit (MWG p. 25). The law of demand says it slopes down; MWG's definition allows the opposite, a Giffen good at (p, w), "not an economic impossibility". Whether that can happen is the next slide, back with slide 19's consumer.

Rice gets cheaper: slide 19's consumer again

Revealed in class

Before class: on slide 19's picture, let rice cost 1.5 instead of 2 and find her new best bundle.

Slide 22 Rice gets cheaper: slide 19's consumer again

Revealed in class.

Before class: on slide 19's picture, let rice cost 1.5 instead of 2, with $w = 12$ and meat at 3, and find her new best bundle. Then compare the amount of rice with the old one, and say what slide 19's wealth zones have to do with it.

Dearer good 2: x_1 falls, stays or rises, depending on σ

gross substitutes: $\partial x_1/\partial p_2 > 0$; gross complements: < 0

CES: $u = \left(\frac{1}{2}x_1^\rho + \frac{1}{2}x_2^\rho\right)^{1/\rho}$, $\sigma = 1/(1-\rho)$; $\sigma = 2$: same $\succsim$ as $\sqrt{x_1} + \sqrt{x_2}$

σ	u	$x_1(p, w)$	p_2: $3 \to 4$ $p_1 = 2$, $w = 12$
0	$\min\{x_1, x_2\}$	$\dfrac{w}{p_1 + p_2}$	$2.4 \to 2$: complements
1	$x_1 x_2$	$\dfrac{w}{2p_1}$	$3 \to 3$: neither
2	$\sqrt{x_1} + \sqrt{x_2}$	$\dfrac{w\, p_2}{p_1(p_1 + p_2)}$	$3.6 \to 4$: substitutes

CES: complements for $\sigma < 1$, neither at $\sigma = 1$, substitutes for $\sigma > 1$.

MWG 3.G, p. 70 ("gross" = Walrasian demand; the Hicksian version is Wednesday). CES: class 3, slide 26.

Slide 23 Dearer good 2: x_1 falls, stays or rises, depending on σ

The cross-price effect with wealth fixed, which MWG calls "gross": good 1 is a gross substitute for good 2 if a dearer good 2 raises x_1, a gross complement if it lowers it. The three rows are class 4's CES family at the same points 2.4, 3, 3.6 as on its slide 29. Leontief buys the goods as a pair, so a dearer pair means less of both: complements. Cobb–Douglas spends $w/2$ on good 1 whatever p_2: neither. At $\sigma = 2$ she leans away from the dearer good: substitutes. In general the sign of $\partial x_1/\partial p_2$ is the sign of $\sigma - 1$.

Gross relations need not run both ways: a dearer good 2 can raise x_1 while a dearer good 1 lowers x_2. MWG's own definition of substitutes uses Hicksian demand, and that net version comes with Hicksian demand on Wednesday 14 October; the net relation is symmetric, which the Slutsky matrix shows next Tuesday. MWG writes weak inequalities; with strict ones a zero effect is "neither".

In pairs: Roy's identity, then the three effects

Find the demand by Roy's identity. Wealth effect? Own price: usual or Giffen? The other price rises: substitutes or complements? Seven minutes.

$v(p, w)$	demand	wealth	own price	other price ↑
$\dfrac{w^3}{p_1 p_2^2}$, find x_1				
$\dfrac{w + p_2}{p_1} - 1 + \ln\dfrac{p_1}{p_2}$, find x_2 class 4's $x_1 + \ln(1 + x_2)$, $p_1 > p_2$				
$\dfrac{w - 5p_1}{p_2 - p_1}$, find x_1 slide 22's consumer, near $p = (2, 3)$, $w = 12$				

Answers revealed in class. Try it in pairs before class.

Slide 24 In pairs: Roy's identity, then the three effects

Try it before class. The recipe: Roy's identity first, minus the price slope over the wealth slope, then three signs: $\partial x/\partial w$ for the wealth effect, the own-price slope for usual or Giffen, the other-price slope for substitutes or complements. Row 2 belongs to a consumer you met in class 4; row 3 to the consumer of slide 19.

Answers revealed in class. Three checks: a demand must come out positive; the denominator is the wealth slope; a cross effect is read off the demand, not off v.

(optional) Good 1 is Giffen: what happens to x_2 when p_1 rises?

A consumer with two goods; Walras' law holds; $\partial x_1/\partial p_1 > 0$.

(1) Show that $\partial x_2/\partial p_1 < 0$.

Hint: slide 20, line 3, with $k = 1$.

(2) Explain (1) with her spending.

(3) Can $\partial x_1/\partial p_2 > 0$ hold too?

At home if not done in class; the answers are in the slides released after class.

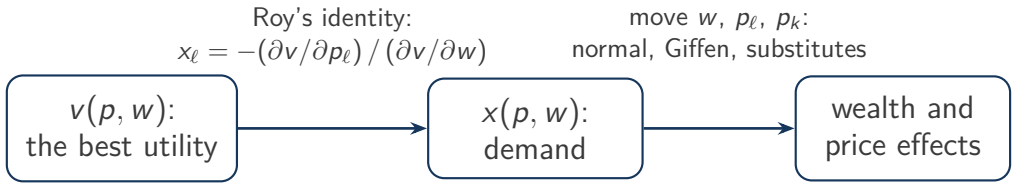
MWG 2.E, pp. 27–28 (Proposition 2.E.2, Cournot aggregation).

Slide 25 (optional) Good 1 is Giffen: what happens to x_2 when p_1 rises?

An optional exercise, to be done at home if class time runs out: two goods, Walras' law, and a good 1 whose demand rises with its own price. The hint on the slide is slide 20's line 3 with $k = 1$: write it out for this case, move the good-1 terms to one side and read off the sign; then say the same thing in words, starting from what she spends on good 1.

The answers are in the slides and handout released after class. For (3), decide first whether the sign of one cross-price effect tells you anything about the sign of the other, and only then look for an example.

Today in one line



Wednesday: expenditure minimisation — Hicksian demand, Shephard's lemma, duality.
Problem set 1: due at the start of class tomorrow, on paper, one copy per group.
Problem set 2 is handed out tomorrow; due in class Wednesday 21 October.

Read MWG 3.E–3.G (or Muñoz-Garcia ch. 2) for Wednesday 14 October.

Slide 26 Today in one line

In one line: $v(p, w)$ is the value of the consumer's problem, homogeneous of degree zero, increasing in wealth, nonincreasing in prices, quasiconvex, and it changes with the utility representation; Roy's identity gets demand back from it with two derivatives, and demand does not change with the representation. Demand moves with wealth (normal or inferior, not everything inferior), with its own price (usually down, Giffen only if inferior) and with the other price (gross substitutes or complements).

Wednesday 14 October: the dual problem, the cheapest way to reach a utility level: Hicksian demand, the expenditure function, Shephard's lemma, duality, and the true cost-of-living index. Reading: MWG 3.E–3.G, or Muñoz-Garcia chapter 2.

Problem set 1 is due at the start of class tomorrow, on paper, one copy per group; problem set 2 is handed out tomorrow and is due in class on Wednesday 21 October.